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COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE:CSA 0611

SLOT:C

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CO4_AT1

1.Find the longest subsequence of a string that is also a palindrome.

Sample Input: s = BBABCBCAB.

Sample Output: Length = 7

CODE:

```
def lps(s):
    n = len(s)
    dp = [[0] * n for _ in range(n)]
    for i in range(n):
        dp[i][i] = 1
    for length in range(2, n + 1):
        for i in range(n - length + 1):
            j = i + length - 1
            if s[i] == s[j]:
                dp[i][j] = 2 if length == 2 else dp[i + 1][j - 1] + 2
            else:
                dp[i][j] = max(dp[i + 1][j], dp[i][j - 1])
    return dp[0][n - 1]
s = input("Enter string: ")
print("Length =", lps(s))
```

OUTPUT:

```
Output

Enter string: BBABCBCAB
Length = 7

=== Code Execution Successful ===
```